

**Databricks, Inc. Comments in Response to
The National Information and Telecommunications Administration
AI Accountability Policy Request for Comment**

Regulations.gov Docket Number NTIA-2023-0005

Databricks, Inc. (Databricks) is pleased to have the opportunity to provide comments in response to the request for comment (RFC) on AI Accountability Policy by the National Telecommunications and Information Administration (NTIA).

Databricks' Vantage Point with Respect to AI

As a privately held company providing services to other businesses rather than to consumers, Databricks is not broadly known. We therefore feel it important to provide some context as to why Databricks has a unique and valuable vantage point regarding the field of artificial intelligence (AI).

Databricks provides a high-performance cloud-based data processing and full life cycle data management platform, optimized, and heavily utilized for AI applications. Databricks has over 5,000 employees, and serves customers in every industry, in all major markets around the globe, on all major cloud infrastructure providers.

Of Databricks' approximately 9,000 enterprise and public sector customers, we estimate that more than 1,000 are building and/or operating AI applications on the Databricks platform. Databricks' customers, including many using its platform for AI, include more than half of the Fortune 500 plus many smaller entities. These organizations use the Databricks platform to design, develop, test and/or host their AI applications because it is optimized for exactly what AI requires most: (1) the ability to process massive amounts of data in a scalable, ultra-high speed manner; (2) the ability to ingest a wide range of data, including unstructured data (the type needed to power image and voice recognition and many other AI applications) in addition to structured data; and (3) the ability to process real-time streaming data in addition to data processed on a batch basis.

Databricks believes sharing of foundational technologies furthers innovation and progress for all, and has extensive experience with the open source software development community. Databricks' founders and employees have led the creation of Apache Spark™, Delta Lake and MLflow - three of the most important and widely used data analytics open source projects of all time. MLflow is a widely used full life cycle AI development and management platform that guides software developers through all the major stages of AI development including planning, writing code, testing, and deployment. The Databricks platform incorporates a very high performance and efficient version of MLflow. Databricks also introduced an open source English-language chatbot called Dolly 2.0 in April 2023, introduced primarily as a proof of concept, but also available for commercial purposes under

the Apache 2.0 license¹. Dolly 2.0 is a large language model (LLM) that follows instructions in English, and was the first such open-source LLM licensed for commercial use.

Databricks customers are using AI systems built and run on the Databricks platform for a wide variety of purposes, with many uses focused on improving the quality of products and services, and improving the experiences of employees and end customers. Just a few examples include:

- Accelerating the pharmaceutical R&D lifecycle for discovery of cures and diagnostics for many illnesses;
- Improving design and management efficiency for telecommunications, energy, transportation and freight networks;
- Improving end-customers' experience in gathering information about products and services; and
- Improving employees' job quality by reducing repetitive, tedious work that can lead to high worker burnout and turnover.

A growing number of these use cases take advantage of open source LLMs like Dolly 2.0 that permit commercial use. Open source LLMs are becoming increasingly important as the basis for AI business enhancement and productivity applications. Many customers use the Databricks platform to retrain or 'tune' the LLM based on technical or other internal documentation specific to their use cases.

As a major provider of AI infrastructure, tools and services to support the development and deployment of open source LLM-based AI applications, Databricks has a uniquely informed view as to the workings of the AI supply chain, and the substantial benefits AI is providing to enterprise and public sector users.

Alignment with Comments Provided by BSA | The Software Alliance

Databricks is a member of BSA | The Software Alliance (BSA). In addition to providing comments on specific issues, we would like to emphasize our support for the key themes of the BSA response to the NTIA RFC:

- **AI Policies on Accountability Should Distinguish Between Developers and Deployers.** Any policies focused on supporting AI accountability should reflect the differences in the roles of developers and deployers and assign obligations accordingly.
- **The AI Audit Ecosystem Is Not Mature Enough to Support Mandatory Third-Party Audits.** The lack of developed standards, procedures, an ecosystem of appropriate auditors, or governing professional bodies makes broad mandatory third-party audits unworkable.
- **AI Regulation is Important and Should Require the Use of AI Impact Assessments for High-Risk Uses of AI Systems.** BSA and Databricks support regulation that requires

¹ The model, including model weights, is licensed under Apache 2.0 while the training data is available under a Creative Commons license (CC-BY-SA) that also permits commercial use. The concept demonstrated was that a highly capable open source LLM usable commercially for 'fine-tuning' for specialized applications (see discussion below) could be created with a relatively low investment in compute time and resources, and using a relatively small training data set.

appropriate parties to conduct impact assessments for high-risk uses of AI systems. By requiring steps to protect against potential risks, regulation will help ensure trust in AI.

- **AI Policies Should be Designed to Permit the Continued Growth and Use of Open Source AI Systems.** Open source AI is an increasingly important part of the AI supply chain, and an area in which Databricks has extensive experience, including the use of ‘fine-tuned’ (retrained for a specialized purpose) open source LLMs in a wide variety of business applications. Through the collaboration of a large community of developers, open source AI provides comparative benefits in other areas, including cost, competition, transparency, accountability, innovation, data security, and quality control. Requiring developers or contributors to an open source project to have knowledge of, or connection to, a downstream participant in the AI supply chain is unrealistic. Any such obligation would effectively ban open source AI and should be avoided within federal AI legislation.

Our Comments Focus on the Important Role Open Source Plays in the AI Supply Chain, and the Need to Ensure Regulation Does Not Effectively Stifle Open Source AI

The fourth of BSA’s key themes, relating to open source AI systems, is one that we want to address in detail because of Databricks’ unique vantage point regarding the development and use of such systems. Open source software is in the public domain and allows developers to inspect and modify the source code that is the foundation of an application. Open source software has made many of today’s leading technologies feasible and more productive, and has been largely responsible for the rapid development of new technologies in recent decades, including the development of AI technologies. The Linux Foundation has estimated *open source constitutes 70% to 90% of any given piece of modern software*². Ensuring that AI regulation does not effectively prohibit open source AI is important because of the many benefits open source development can provide to businesses, consumers and society as a whole. As context for many of our later comments regarding open source AI, we feel it is important to provide some elaboration on these benefits, as follows:

- **Collaboration and Innovation** – The collaborative aspect of open source spurs innovation, knowledge sharing, collective problem-solving, and the pooling of resources, which can accelerate innovation, improve AI capabilities, and lead to more robust and secure systems. Enhanced collaboration also enables a greater diversity of input into model design, model training, and ongoing quality monitoring and enhancement.
- **Lower Cost, More Competition and ‘Democratization’ of AI** – LLMs are expensive to operate. Using an open source LLM brings the operating cost down considerably by eliminating licensing and access fees. Open source lowers barriers to entry, improving competition, cost of ownership, and the range of alternatives, thus ‘democratizing’ the use and control of AI. Open source reduces vendor lock-in, giving users greater flexibility and ability to control their AI system, and to negotiate more favorable pricing. Open source AI helps counterbalance the concentration of AI in the hands of a few large and influential technology companies that provide closed AI models.
- **Control of Data and Improved Data Governance/Security** – Data governance is easier to manage when the data does not leave the user’s control. Because open source code can be brought into an enterprise’s own model, it does not require moving data the way third-party closed AI models do. With open source a user can have full ownership and control of the data used

² <https://www.linuxfoundation.org/blog/blog/a-summary-of-census-ii-open-source-software-application-libraries-the-world-depends-on>

to train the AI model, with no need to share new data to ‘tune’ the model, and less worries about control and security of sensitive data.

- **Quality/Latency** – With open source, a community of contributors identifies and resolves issues, providing what is effectively ‘crowd-sourced’ quality control. This aspect of open source enables quicker mitigation of issues relating to accuracy or other harmful content. In addition, with open source, the user can fully integrate an LLM with the user’s business logic which can reduce latency, something generally impossible to do with closed LLMs.
- **Customization/Flexibility** – With open source, the user has the flexibility to create a customized application using its own data that it controls, and no need to depend on an LLM vendor for customization. The AI system can be self-hosted or hosted by a vendor at the user’s discretion – the user is not reliant on API access to a vendor platform or on third party service level agreements.
- **Transparency/Accountability** – Organizations want to understand exactly what is happening in their models, and open source makes that possible because the source code - and typically the model weights and training data - are freely available for access and inspection, not only to the user but to any third party that may wish to perform an analysis, including regulators and others with concerns about how the AI system works. Open source accessibility and transparency enable more rapid identification and mitigation of problems that may arise, permit greater understanding of how the AI system works, and generally enhance trust in AI. Regulators are understandably looking for this kind of transparency with respect to high risk AI systems. With open source, all systems, whether low risk or high risk, provide complete transparency from the moment of release.

Databricks’ Responses to Specific Questions:

(Note: Databricks is providing feedback only where we feel we provide a unique perspective, as discussed above, as opposed to areas where we feel our feedback would be consistent with extensive feedback you will receive from numerous other parties).

Question 5. *Given the likely integration of generative AI tools such as large language models (e.g., ChatGPT) or other general-purpose AI or foundational models into downstream products, how can AI accountability mechanisms inform people about how such tools are operating and/or whether the tools comply with standards for trustworthy AI?*

Transparency is an important goal in promoting AI accountability. As policy approaches to enhance transparency are developed, it is important to distinguish between the different roles that companies³ play in the AI ecosystem. Developers design, code, produce or make material modifications to AI systems. In contrast, deployers actually implement AI systems in specific use cases⁴. While developers are best positioned to understand the intended uses and limitations of an AI system, deployers typically interact with consumers and decide when and how to deploy the AI system for a particular use. Developers and deployers have distinct roles to play in ensuring appropriate education and transparency for consumers. With respect to open source foundation models used in AI applications,

³ References to ‘companies’ in this response may include, where appropriate, governmental entities, research organizations or other non-business entities.

⁴ In some cases, a party may engage in both the role of developer and deployer, for example where a deployer makes material modifications to an AI system prior to deploying it, or when a developer creates an AI system for its own internal use.

any obligation to inform people about how such tools are operating should rest with the party placing the AI application into actual use (i.e., the deployer). In this scenario, the deployer is the party exposing people to the application and creating the potential risk.

It is important to note that AI accountability is heavily context dependent. Developing an accountability framework based on a single use case — i.e., consumer-facing uses — can be detrimental to many applications used in other contexts, such as business-to-business settings that may present vastly different expectations and risk considerations. As an example, many companies are building internal business tools that enhance products and services using open source LLMs as a base and fine-tuning (retraining) the open source LLM using technical information specific to their customer's internal use case. This application of the original open source LLM presents significantly fewer potential risks than would a consumer-facing application of an LLM, with the consumer-facing application potentially warranting a more rigorous accountability framework applicable to the deployer.

Question 7. *Are there ways in which accountability mechanisms are unlikely to further, and might even frustrate, the development of trustworthy AI? Are there accountability mechanisms that unduly impact AI innovation and the competitiveness of U.S. developers?*

With respect to open source AI systems, any aspect of an accountability mechanism that would require an originator of, or contributor to, an open source AI system to monitor or interact with downstream users, or to have specific knowledge of how the open source model is eventually deployed by third parties, would be unworkable because open source software is created to reside in the public domain to be used by anyone, resulting in the full transparency that is an inherent attribute of open source software. Any such accountability requirement would effectively prohibit the development of such open source AI systems, eliminating the substantial benefits provided by the availability of open source AI, including furthering innovation and keeping the cost of productivity-enhancing AI low for a wide range of businesses.

Question 15. *The AI value or supply chain is complex, often involving open source and proprietary products and downstream applications that are quite different from what AI system developers may initially have contemplated. Moreover, training data for AI systems may be acquired from multiple sources, including from the customer using the technology. Problems in AI systems may arise downstream at the deployment or customization stage or upstream during model development and data training.*

- a. Where in the value chain should accountability efforts focus?*
- b. How can accountability efforts at different points in the value chain best be coordinated and communicated?*
- c. How should vendors work with customers to perform AI audits and/or assessments? What is the role of audits or assessments in the commercial and/or public procurement process? Are there specific practices that would facilitate credible audits (e.g., liability waivers)?*
- d. Since the effects and performance of an AI system will depend on the context in which it is deployed, how can accountability measures accommodate unknowns about ultimate downstream implementation?*

Broadly speaking, as discussed in our response to question 5, the AI supply chain includes developers (parties that design and build AI systems) and deployers (parties that place an AI system into use). Any policies creating obligations applicable to participants in the AI supply chain should reflect these different roles and assign obligations accordingly.

Communication among the different AI actors is important to ensuring the successful operation of accountability frameworks across the lifecycle of an AI system. Developers that design an AI system should have obligations to provide deployers using that AI system with the information reasonably necessary for the deployer to conduct an impact assessment. This may include the AI system's capabilities, known limitations, and guidelines for intended use. With this information, a deployer can assess how they might use an AI system in light of the developer's intended use for the system and its known limitations. At the same time, because developers will not have insight into the actual use of the AI system, a deployer should be responsible for monitoring issues that arise in downstream implementation, including having a feedback mechanism to surface issues that may arise.

An important and growing aspect of the AI supply chain consists of the use of open source AI models and components, with this growth accelerating due to the recent introduction of highly capable open source LLMs that permit commercial use. Following the release of ChatGPT, much attention has been focused on proprietary LLM innovations. At the same time, the open source community has helped develop LLMs that address many of the challenges posed by proprietary LLMs⁵, thereby allowing more enterprise users to realize the many benefits enabled by LLMs (discussed further below). Many companies are now building internal business tools using open source LLMs as a base, fine-tuning (retraining) the open source LLM with technical information specific to their customer's internal use case. This application of the original open source LLM presents significantly fewer potential risks than would a consumer-facing application of an LLM. Fine-tuned open source LLMs are being used by businesses and governments for a wide range of purposes, in most cases to improve the quality of products or services, or to enhance the experiences of users or employees.

Given these benefits, and the benefits of open source AI previously discussed, AI regulation should not serve as a roadblock to open source AI. As discussed above, it is desirable that AI regulation make a distinction between the developer of AI (the party designing and engineering the AI system), and a deployer (the party employing the AI system in a particular use case). Differing regulatory obligations should apply to developers and deployers, with developers' obligations relating primarily to steps taken during the design and development stage prior to deployment, and deployers' obligations relating to how the AI system is used at launch and thereafter. In the context of open source AI, it is important that regulation take into account the fact that an open source developer has no way of monitoring who eventually uses the open source AI, and no way to monitor how it is used. Any obligations on the open source developer should be limited to what the developer does prior to, and up to the point of, releasing the system as an open source project, with the obligations focused on design steps, pre-release testing, risk assessment and calibration, and documentation through the point of release. Any regulatory obligations applied subsequent to release should be borne by the deployer. It should be made clear in the regulations that the act of releasing a project as open source

⁵ Examples of these challenges include high retraining and operating costs, limited flexibility as to customization, concerns around protection and control of sensitive data used in retraining or within inputs to the vendor's API-based application, lack of code transparency for use in analyzing security, performance or modification issues, and potential vendor lock-in.

(or making a contribution to an existing open source project) does not, by itself, constitute a ‘deployment’. For purposes of regulation, the deployer of an open source software AI system should be considered to be the party that actually puts such a system into use.

Regulation of open source AI that creates post-release obligations on the developer, including any obligation to interact with downstream players or have ongoing specific knowledge of how the open source model is eventually deployed by third parties, would be unworkable because of the fundamental nature of open source. Regulation along these lines would effectively quash open source AI, guaranteeing that powerful AI systems will be the domain of a few large companies deploying ‘black box’ systems. This would heighten the concentration of power over AI in the hands of a few, which would in turn limit scrutiny and research critical to the public’s understanding of AI and hinder the goal of creating public trust in AI.

Because of their advantages and growing popularity, highly capable open source LLMs are seen as a potential competitive threat to the highly capable proprietary LLMs provided by a small number of very large technology companies. The following are examples of recent articles highlighting this view:

- <https://bdtechtalks.com/2023/05/08/open-source-llms-moats/>
- <https://www.kdnuggets.com/2023/05/llm-apocalypse-revenge-open-source-clones.html>
- <https://www.thestack.technology/open-source-chatgpt-alternatives/>
- <https://bootcamp.uxdesign.cc/seizing-the-open-source-ai-opportunity-how-google-and-openai-could-lose-ground-a91411458880>

Databricks is concerned that the large technology companies offering powerful proprietary LLMs will encourage broadly applicable regulation of LLMs, including ongoing downstream obligations, that such large players can easily meet based on the nature of proprietary models, but that would effectively take open source LLM competition off the table. It is important that creators of public policy relating to AI take this dynamic into consideration.

Question 16. *The lifecycle of any given AI system or component also presents distinct junctures for assessment, audit, and other measures. For example, in the case of bias, it has been shown that ‘[b]ias is prevalent in the assumptions about which data should be used, what AI models should be developed, where the AI system should be placed—or if AI is required at all.’ How should AI accountability mechanisms consider the AI lifecycle? [sub questions omitted here]*

In the context of open source AI, it is important that regulation take into account the fact that an open source developer has no way of monitoring who eventually uses the open source AI, and no way to monitor how it is used. Any obligations on the open source developer, including impact assessments relating to high risk AI systems, should be limited to what the developer does prior to, and up to the point of, releasing the system as an open source project. Any regulatory obligations applied subsequent to release should be borne by the deployer. Regulation of open source AI that creates post-release obligations on the developer, including any obligation to interact with downstream players or have ongoing specific knowledge of how the open source model is eventually deployed by third parties, would be unworkable because of the fundamental nature of open source, and will effectively quash open source AI. In addition to this workability issue, relative to allocation of regulatory burden, it is important to note that developers of open source software are not directly compensated for their

development efforts, whereas the deployer typically is directly compensated, or otherwise benefits, from use of the open source software, making it more appropriate to place the burden on the deployer.

Question 20. *What sorts of records (e.g., logs, versions, model selection, data selection) and other documentation should developers and deployers of AI systems keep in order to support AI accountability. How long should this documentation be retained? Are there design principles (including technical design) for AI systems that would foster accountability-by-design?*

In keeping with our prior comments, in the context of open source AI, it is important that any documentation requirements imposed on the developer of, or contributor to, an open source AI system or component be limited to material created, and actions taken, by such party as of the time of release as, or contribution to, open source. Any required documentation can be submitted to the appropriate open source repository, where it will be accessible to downstream players accessing and utilizing the open source material.

Question 21. *What are the obstacles to the flow of information necessary for AI accountability either within an organization or to outside examiners? What policies might ease researcher and other third-party access to inputs necessary to conduct AI audits or assessments?*

One aspect of open source AI relevant to accountability and transparency is that the model source code - and typically the model weights and training data - are freely available for access and inspection, not only to the user but to any third party that may wish to perform an analysis, including regulators and others with concerns about how the AI system works. This accessibility and transparency will facilitate assessments when needed, enable more rapid identification and mitigation of problems that may arise, permit greater understanding of how the AI system works, and generally enhance trust in AI. With open source, all systems, whether low risk or high risk, provide this complete transparency from the moment of release.

Question 26. *Is the lack of a federal law focused on AI systems a barrier to effective AI accountability?*

A federal law focused on AI systems would create an important mechanism to promote effective AI accountability. Without a federal law, responsible companies will continue to examine and address AI risks, but this will not be the case for all AI actors. In addition, a patchwork of state and local laws could emerge, creating differing obligations for the same AI systems and potentially resulting in stifled innovation.

Question 34. *Is it important that there be uniformity of AI accountability requirements and/or practices across the United States? Across global jurisdictions? If so, is it important only within a sector or across sectors? What is the best way to achieve it? Alternatively, is harmonization or interoperability sufficient and what is the best way to achieve that?*

Uniform legal requirements for AI systems will be important to foster responsible development in these technologies. In the United States, there should be a national law with federal preemption that applies consistent standards throughout the country. As more countries seek to regulate AI, global interoperability across legal requirements applied to AI systems will become increasingly important so that companies that develop and deploy AI systems can create strong compliance programs that

meet legal obligations across the countries in which they operate. As a result, there is a need for consistent rules on AI accountability to ensure that companies identify and mitigate potential risks involved in an AI system across its lifecycle.

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Thank you for the opportunity to provide comments on AI accountability. Databricks looks forward to serving as a potential resource as you continue to engage in policy discussions on this issue.